AI ASSISTED CODING

LAB-4

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BATCH:13

1. Palindrome Number — Zero-Shot Prompting

Objective: To use zero-shot prompting to generate a Python program that checks whether a given number is a palindrome, without providing any examples, and to evaluate the generated code using multiple test cases and edge cases.

CODE:

```python
def is_palindrome(number):
    if number < 0:
        return False

    original = number
    reverse = 0

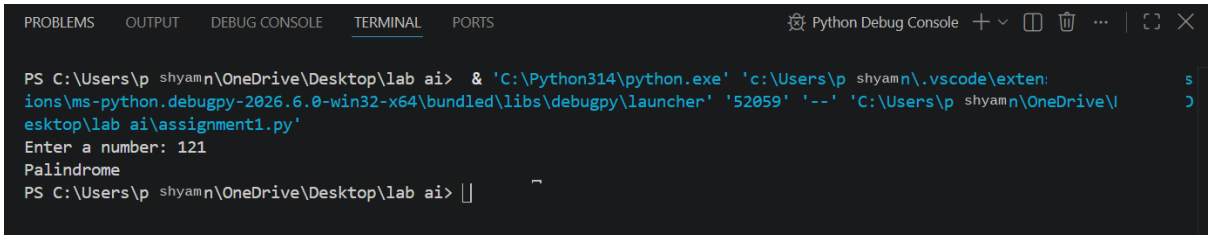
    while number > 0:
        digit = number % 10
        reverse = reverse * 10 + digit
        number //= 10

    return original == reverse


num = int(input("Enter a number: "))

if is_palindrome(num):
    print("Palindrome")
else:
    print("Not a palindrome")
```

OUTPUT:



2. Factorial Calculation — One-Shot Prompting

Objective: To use one-shot prompting by providing a single input-output example to generate a Python program for calculating the factorial of a given number, and to analyze how the example influences the clarity and correctness of the generated code.

CODE:

```python
def factorial(n):
    result = 1

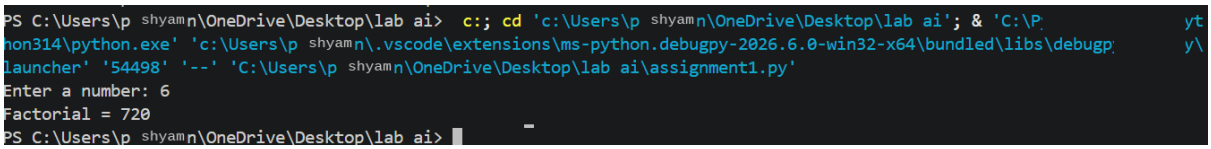
    for i in range(1, n + 1):
        result = result * i

    return result


num = int(input("Enter a number: "))

if num < 0:
    print("Factorial is not defined for negative numbers.")
else:
    print("Factorial =", factorial(num))
```

OUTPUT:



3. Armstrong Number — Few-Shot Prompting

Objective: To use few-shot prompting with multiple input-output examples to generate a Python program that checks whether a given number is an Armstrong number, and to evaluate the influence of multiple examples on the accuracy and structure of the generated code.

CODE:

```python
def is_armstrong(number):
    digits = str(number)
    power = len(digits)

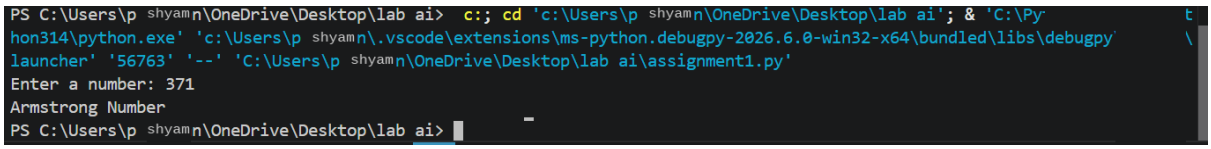
    total = sum(int(digit) ** power for digit in digits)

    return total == number


num = int(input("Enter a number: "))

if is_armstrong(num):
    print("Armstrong Number")
else:
    print("Not an Armstrong Number")
```

OUTPUT:



4. Perfect Number — Zero-Shot Prompting

Objective: To use zero-shot prompting to generate a Python program that checks whether a given number is a perfect number, and to identify missing conditions, edge cases, and possible inefficiencies in the AI-generated solution.

CODE:

```python
def is_perfect(number):
    if number <= 1:
        return False

    divisor_sum = 0

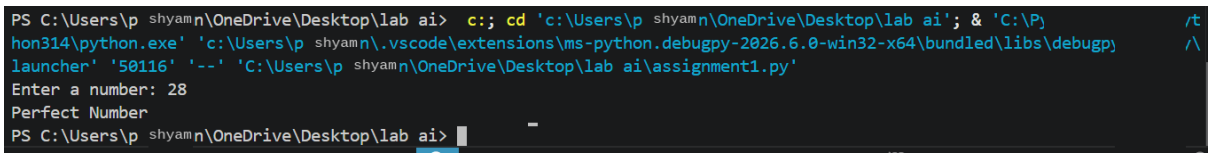
    for i in range(1, number):
        if number % i == 0:
            divisor_sum += i

    return divisor_sum == number


num = int(input("Enter a number: "))

if is_perfect(num):
    print("Perfect Number")
else:
    print("Not a Perfect Number")
```

OUTPUT:



5. Even or Odd Classification — Few-Shot Prompting

Objective: To use few-shot prompting with multiple input-output examples to generate a Python program that classifies a number as even or odd, while incorporating proper input validation and testing the program with negative and non-integer inputs.

CODE:

```
en_odd.py
def check_even_odd(number):
    if number % 2 == 0:
        return "Even"
    else:
        return "Odd"


try:
    num = int(input("Enter an integer: "))
    print(check_even_odd(num))

except ValueError:
    print("Invalid input. Please enter an integer.")
```

OUTPUT:

